IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Elven Jr. | IBM Data Science Capstone | October 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Objective:** Predict Falcon 9 first-stage landing outcomes from historical launch records.
- **Analysis:** 90 launches for EDA and modeling, 101 for SQL, and 56 for the dashboard.
- **Methods:** SQL and visual exploration, Folium maps, Plotly Dash, and four classifiers.
- **Results:** All models reached 83.33% test accuracy. Decision Tree led cross-validation at 87.68%.
- **Interpretation:** All four models made identical predictions on only 18 test launches.

Introduction

- **Context:** First-stage recovery supports the reuse of Falcon 9 boosters.
- **Problem:** Estimate whether a launch will have a successful landing outcome.
- **Research question:** How do payload, orbit, launch site and flight history relate to success?
- **Scope:** Explore historical IBM course datasets covering launches through 2020.
- **Use:** Compare launch conditions and evaluate predictive models for decision support.

https://github.com/elvenj/elvenj-ibm-data-science-professional-certificate



Section
1

Methodology

Methodology Overview

- **Data collection:** SpaceX JSON and Wikipedia HTML exercises, with prepared IBM datasets for analysis.
- **Data wrangling:** Inspect missing values, define the target and encode categorical features.
- **Exploratory analysis:** Compare launch conditions using Pandas, charts and SQLite queries.
- **Interactive analytics:** Map launch sites with Folium and filter outcomes in Plotly Dash.
- **Predictive modeling:** Tune Logistic Regression, SVM, Decision Tree and KNN with 10-fold GridSearchCV.

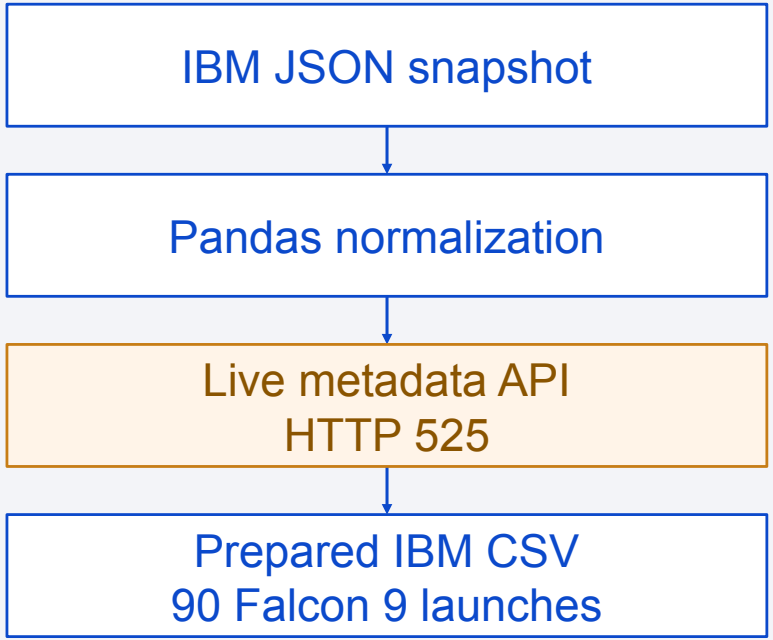
Data Collection

- **API exercise:** Retrieve the IBM-hosted SpaceX JSON snapshot and normalize launch records.
- **Scraping exercise:** Parse Wikipedia launch tables with BeautifulSoup. Row extraction encountered an error.
- **Analysis input:** Use the prepared IBM CSV files to continue wrangling and analysis.



Data Collection: SpaceX API

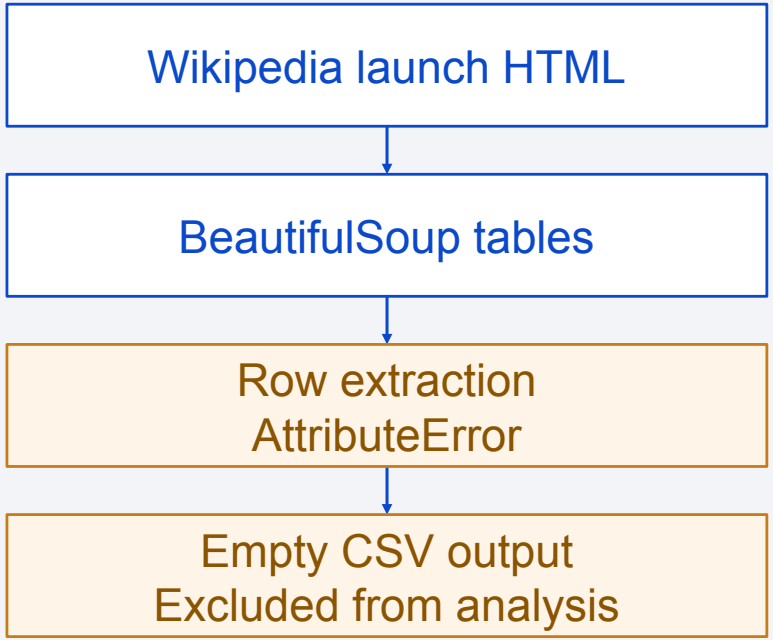
- The IBM JSON snapshot returned HTTP 200.
- Requests and Pandas normalized the launch records.
- Live metadata endpoints returned HTTP 525 and blocked enrichment.
- The analysis continued with 90 prepared Falcon 9 records.



Notebook: SpaceX API collection

Data Collection: Web Scraping

- Retrieve the Wikipedia launch-list HTML.
- BeautifulSoup finds launch tables and their column headers.
- The row loop extracts payload, orbit, site and outcome fields.
- A NoneType error interrupted extraction. The published CSV has zero rows.

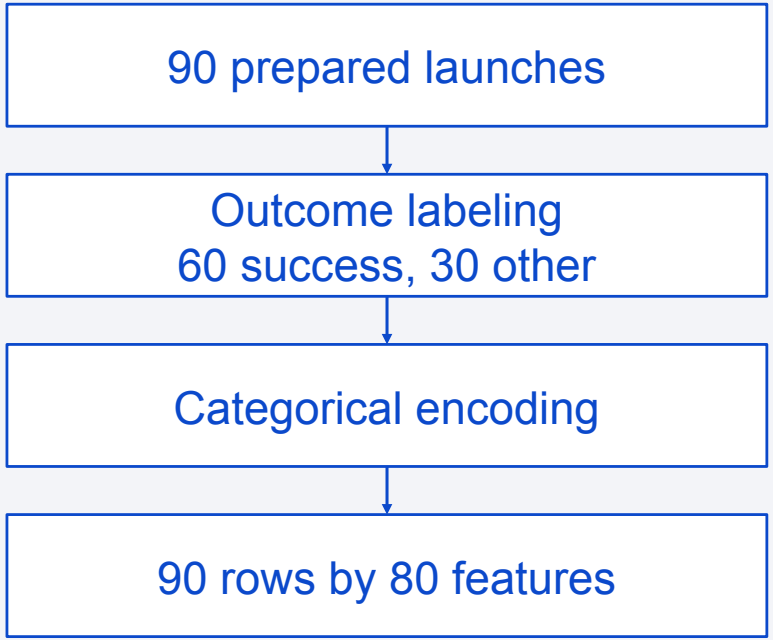


Notebook: Wikipedia web scraping

Data Wrangling

- Retain 90 Falcon 9 launch records.
- Inspect data types and missing values. LandingPad has 26 null values.
- Encode True outcomes as Class 1 and false or undefined outcomes as Class 0.
- One-hot encode categories to produce 80 numeric predictors.

Notebooks: data wrangling and EDA



EDA with Data Visualization

- **Scatter plots:** Flight number and payload versus launch site and orbit show where success and failure occur.
- **Orbit chart:** Mean Class estimates the observed success rate for each orbit.
- **Yearly trend:** Annual mean Class tracks changes in recorded outcomes over time.
- **Feature preparation:** Encode orbit, launch site, landing pad and booster serial with Pandas get_dummies.
- **Scope:** 90 launches across three site labels. Associations support exploration, not causal conclusions.

Notebook: EDA with visualization

EDA with SQL

- **Database:** Load 101 launch records into a SQLite table.
- **Filters:** Find launch sites, customer subsets, booster versions and date ranges.
- **Aggregations:** Use SUM, AVG, MIN, MAX and grouped counts to summarize missions.
- **Comparisons:** Inspect drone-ship outcomes, maximum payloads and the 2015 failures.
- **Data audit:** Trim outcome labels and compare exact-text filters with family-prefix filters.

Notebook: EDA with SQLite

Interactive Map with Folium

- **Location markers:** Show four launch-site labels in the 56-record geospatial dataset.
- **Outcome markers:** Green represents Class 1 and red represents Class 0.
- **MarkerCluster:** Group overlapping launch records for interactive exploration.
- **Distance line:** Connect CCAFS LC-40 to a selected coastline point using a Haversine distance.
- **Map controls:** Use zoom, popups and MousePosition to inspect coordinates.

Notebook: Folium map exercise

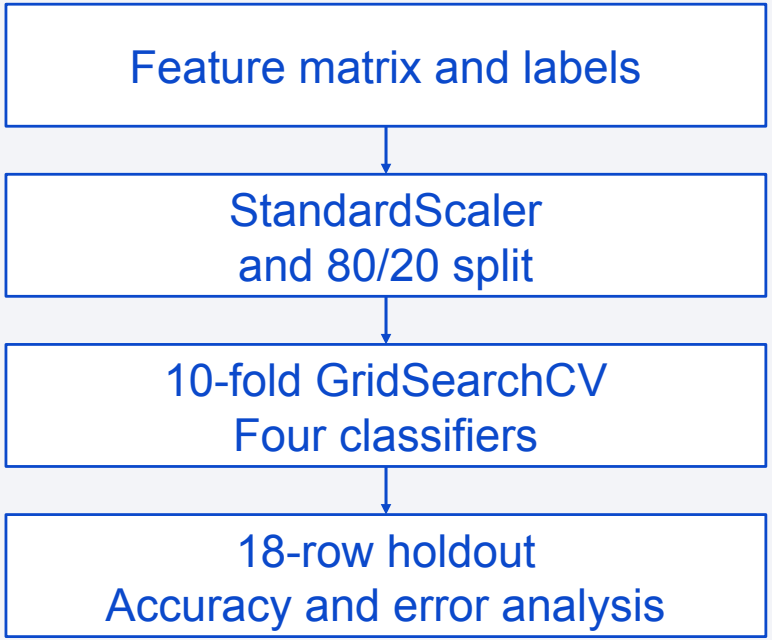
Dashboard with Plotly Dash

- **Site dropdown:** Choose All Sites or one of four site labels.
- **Pie chart:** All Sites compares successful-launch counts. A selected site shows success versus failure.
- **Payload slider:** Filter the scatter plot over a selected payload range.
- **Scatter plot:** Compare payload with binary outcomes, colored by booster version category.
- **Callbacks:** Connect the site and payload inputs to the chart component IDs.

Python app: SpaceX dashboard

Predictive Analysis: Classification

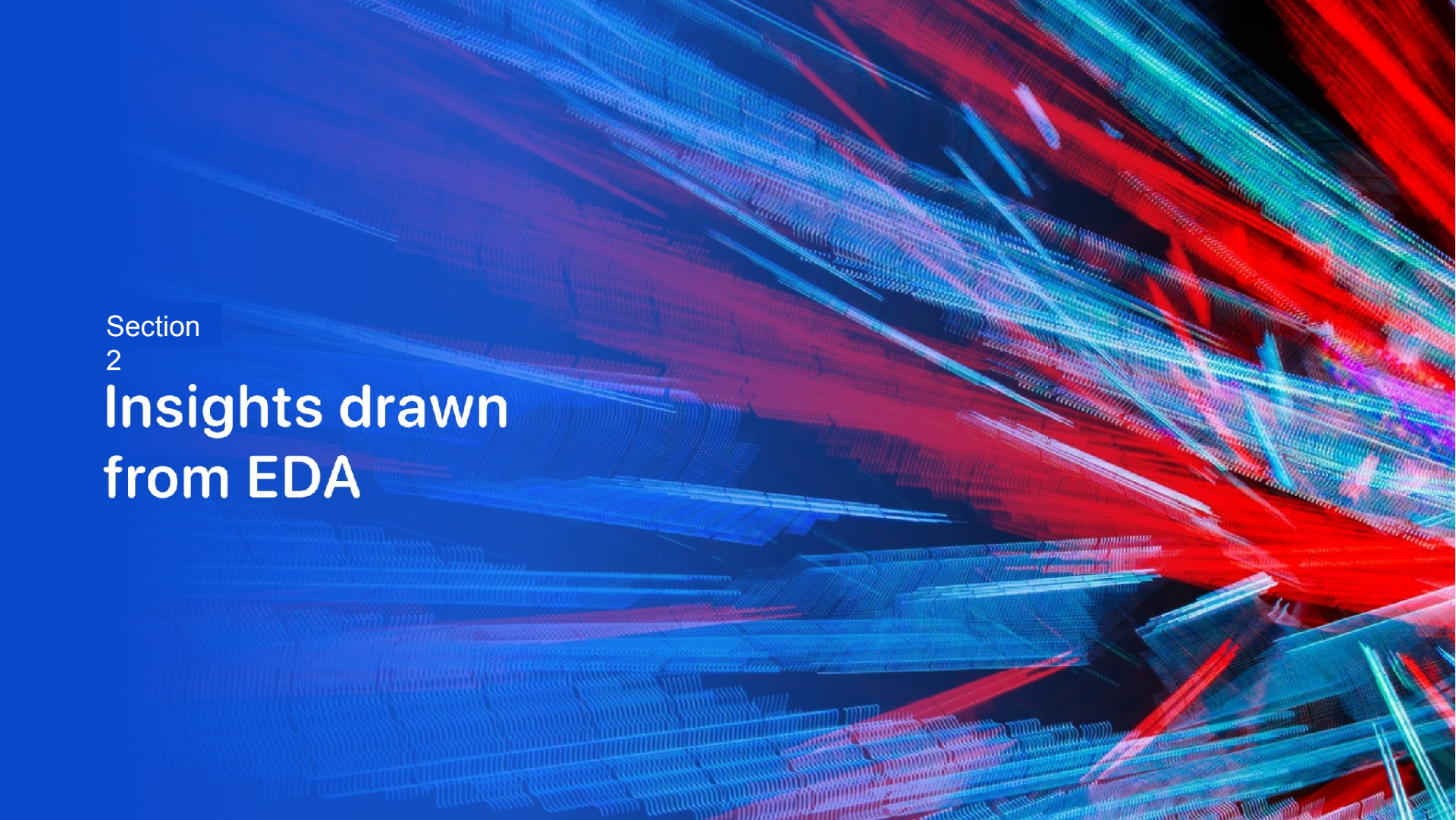
- Use 90 labeled rows with 80 numeric features.
- Standardize predictors, then split 72 train and 18 test rows with random_state=2.
- Tune four models with 10-fold GridSearchCV.
- Compare CV accuracy, test accuracy and confusion matrices.



Notebook: machine learning prediction

Results

- EDA: observed success rates differ by site, orbit and year.
- SQL: launch history, payload summaries and outcome rankings.
- Interactive analytics: launch locations, dashboard filters and payload comparisons.
- Prediction: four models share 83.33% accuracy on the 18-row test set.

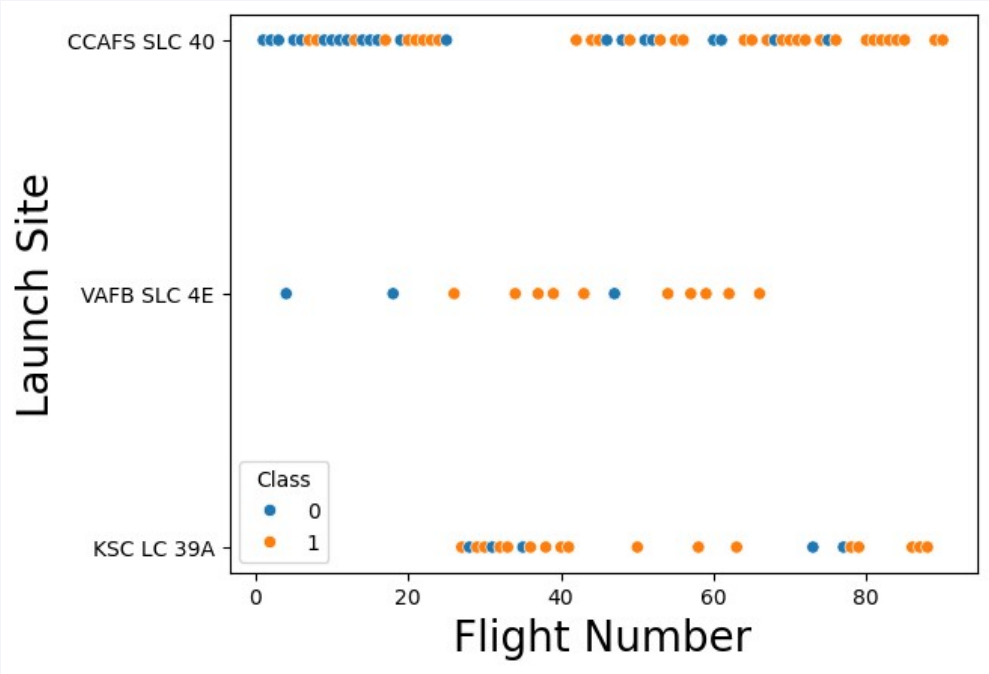


Section 2

Insights drawn from EDA

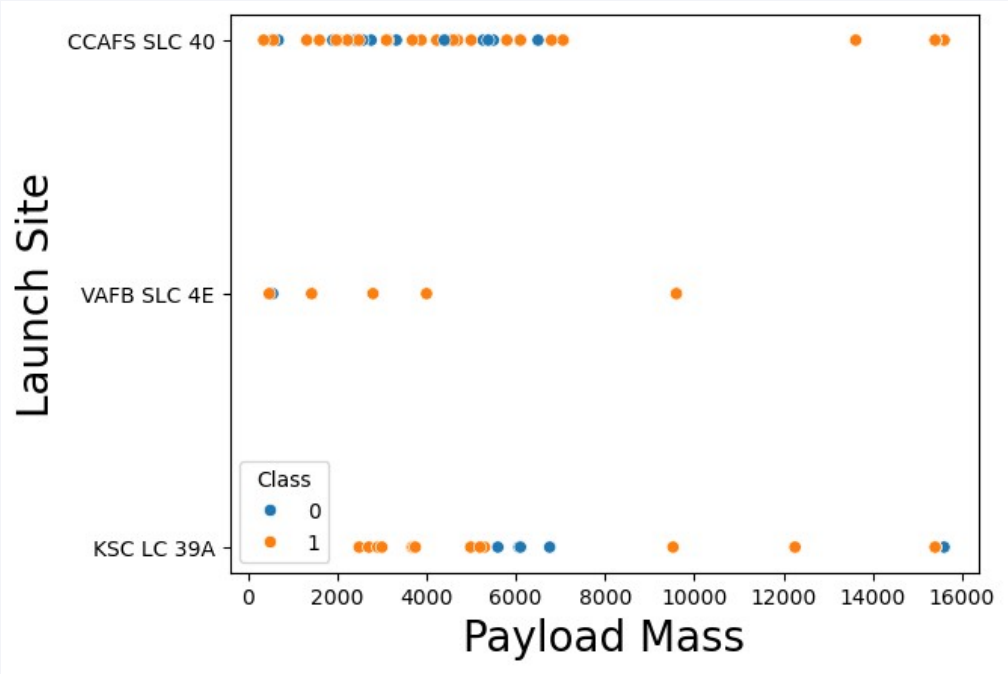
Flight Number vs. Launch Site

- CCAFS SLC 40 has the most launches: 55 of 90.
- KSC LC 39A records 17 successes in 22 launches.
- Later flight numbers contain more successful outcomes.



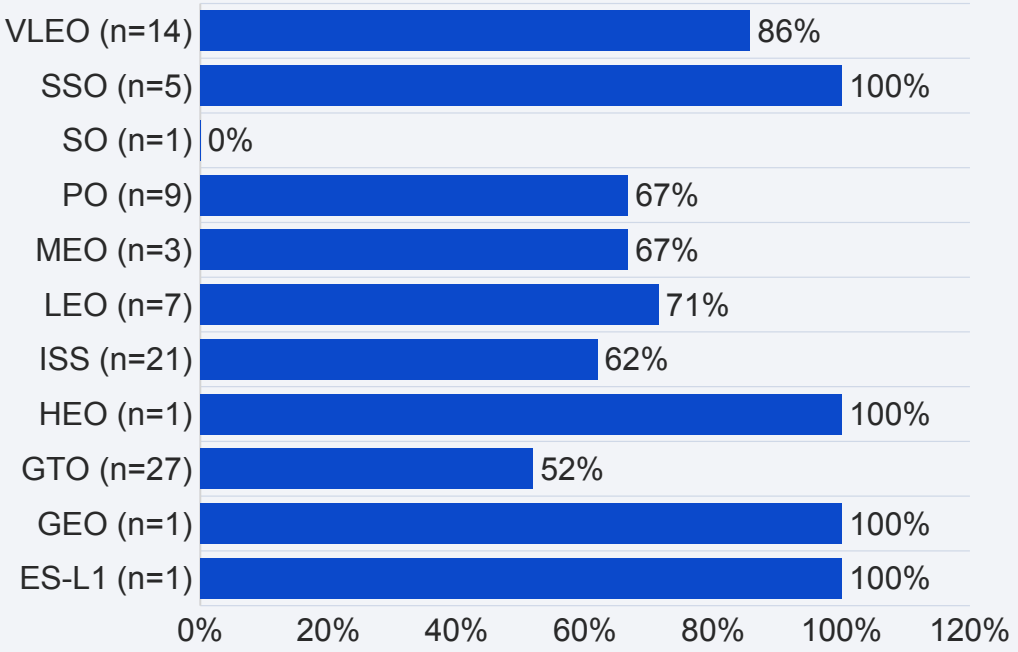
Payload vs. Launch Site

- Success and failure overlap across payload ranges.
- VAFB SLC 4E has no launch above 10,000 kg in this sample.
- Payload alone does not separate outcomes.



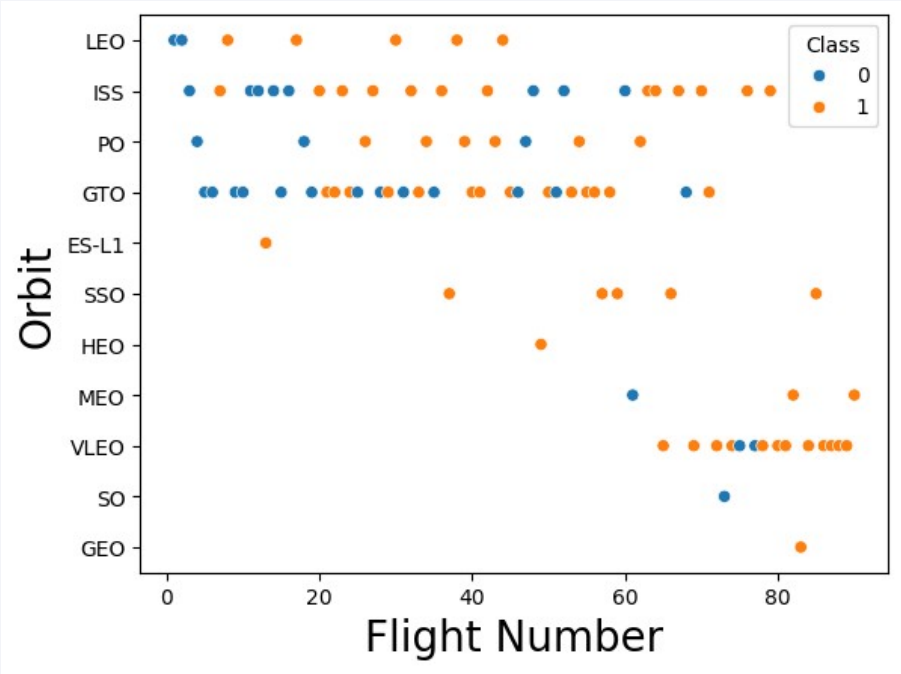
Success Rate vs. Orbit Type

- SSO has 5 successes in 5 launches.
- GTO records 14 successes in 27 launches, about 51.9%.
- Three 100% orbit rates rely on one launch each.



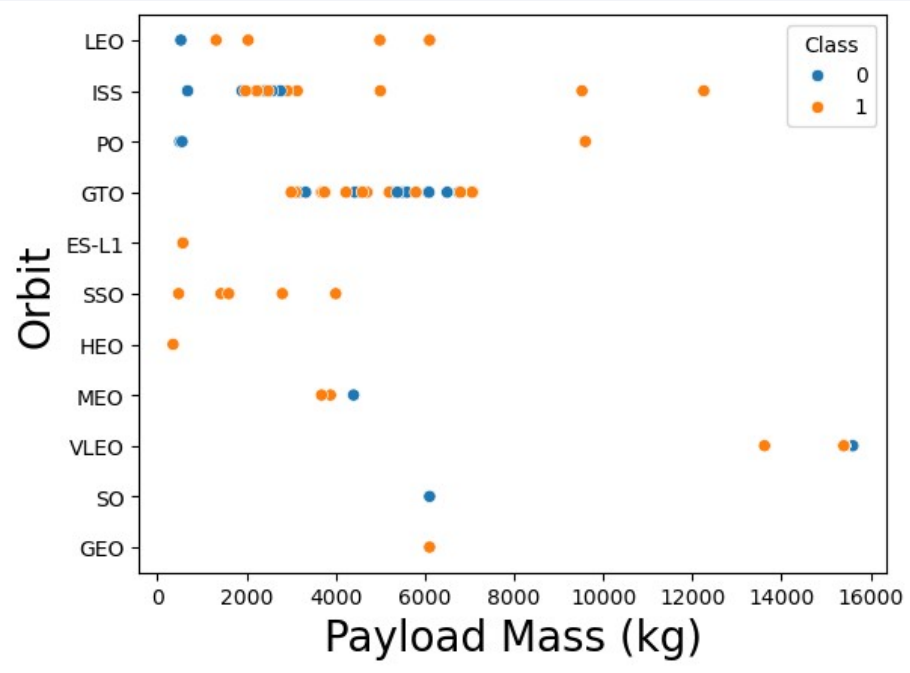
Flight Number vs. Orbit Type

- GTO and ISS have the most records: 27 and 21 launches.
- Later VLEO flights are mostly successful.
- Flight number also reflects time and changing technology.



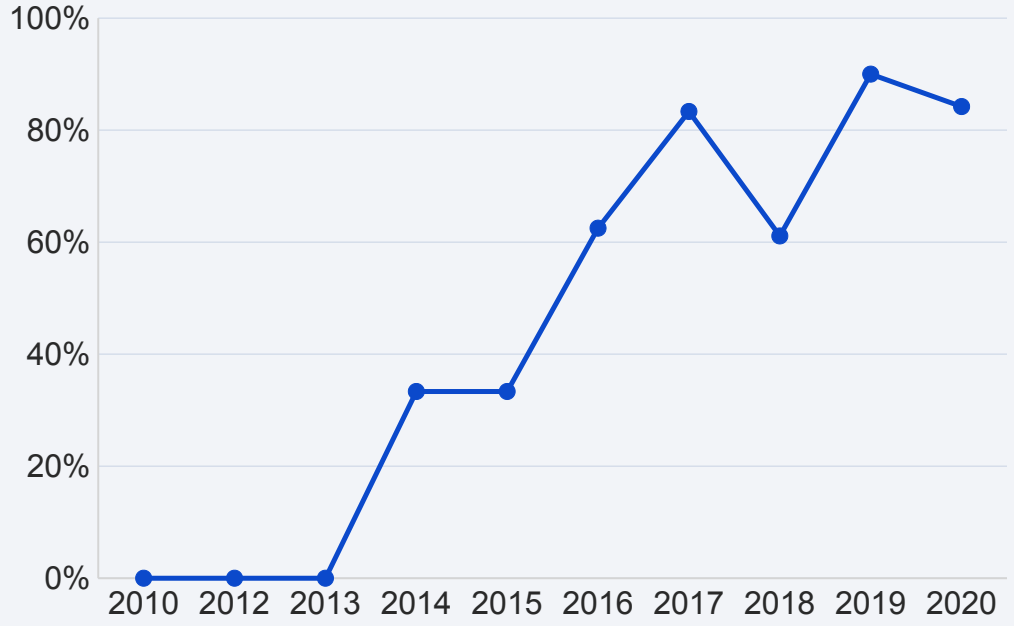
Payload vs. Orbit Type

- VLEO includes the heaviest payloads in the modeling dataset.
- GTO outcomes overlap at similar payload masses.
- Payload ranges depend on orbit and mission requirements.



Launch Success Yearly Trend

- The observed rate rises from 0% in early years to 90% in 2019.
- 2020 records 16 successes in 19 launches, about 84.2%.
- The trend includes dips and changes in the launch mix.



All Launch Site Names

SELECT DISTINCT Launch_Site
FROM SPACEXTBL;

- The SQL dataset contains four distinct text labels.
- CCAFS LC-40 and CCAFS SLC-40 are separate labels in this snapshot.

Launch site label
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Sites Beginning with 'CCA'

WHERE Launch_Site LIKE 'CCA%'
LIMIT 5

Date	Booster version	Site	Payload kg
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677

Total Payload Mass: NASA Filters

SUM(PAYLOAD_MASS__KG_)
WHERE Customer = 'NASA (CRS)'

Customer filter	Payload mass
Exact 'NASA (CRS)'	45,596 kg
Any label containing 'NASA'	107,010 kg

The published query covers CRS only. The broader check also includes shared-customer missions.

Average Payload Mass: F9 v1.1

AVG(PAYLOAD_MASS__KG_)
WHERE Booster_Version = 'F9 v1.1'

Booster filter	Rows	Average payload
Exact 'F9 v1.1'	5	2,928.40 kg
Prefix 'F9 v1.1%'	15	2,534.67 kg

The prefix check includes version labels with booster serial numbers. Query scope changes the average.

First Successful Ground Landing

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)'
```

First recorded date
2015-12-22

The database records the first successful ground-pad landing on 22 December 2015.

Drone Ship Landings: 4,000-6,000 kg

```
Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000
```

Booster version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Four booster-version records match.
- The range excludes both endpoints.

Mission Outcomes

```
SELECT TRIM(Mission_Outcome), COUNT(*)
FROM SPACEXTBL
GROUP BY TRIM(Mission_Outcome);
```

Mission outcome	Count
Success	99
Success (payload status unclear)	1
Failure (in flight)	1
Total	101

TRIM merges the original Success and Success-with-space labels. Mission success differs from booster landing.

Boosters Carrying Maximum Payload

Maximum recorded payload: 15,600 kg. Twelve matching booster-version entries.

Booster version	Booster version
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

Drone Ship Landing Failures in 2015

Year = 2015 and Landing_Outcome = 'Failure (drone ship)'

Month	Booster version	Launch site
January	F9 v1.1 B1012	CCAFS LC-40
April	F9 v1.1 B1015	CCAFS LC-40

Both matching records launched from CCAFS LC-40.

Landing Outcomes: 2010-2017

Date range: 2010-06-04 to 2017-03-20, inclusive.

Landing outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

No attempt is the most frequent label. Outcomes retain distinct landing modes.



Section 3

Launch Sites Proximities Analysis

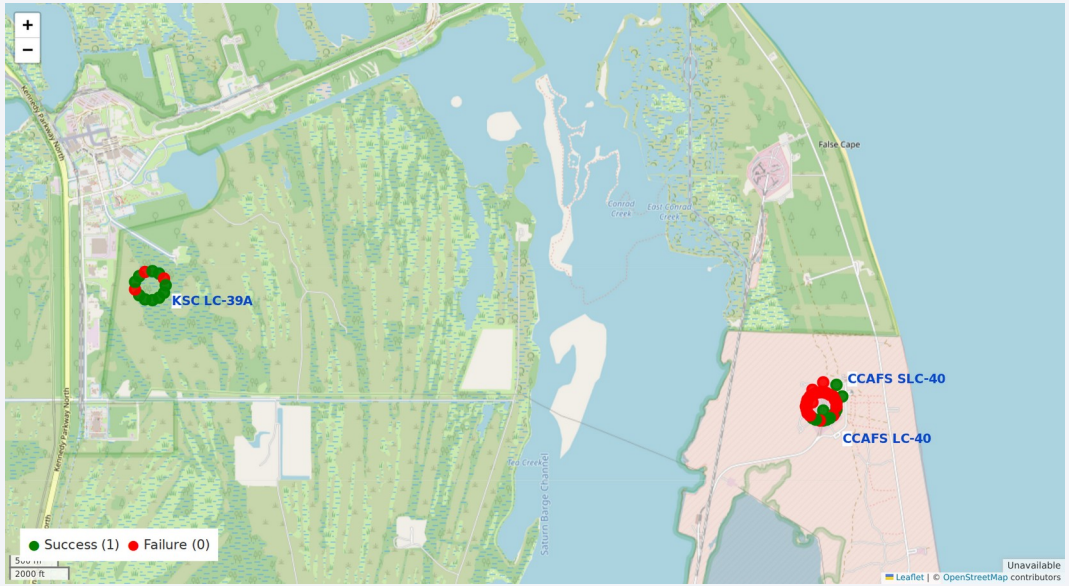
Launch Site Locations

- Four site labels span Florida and California.
- CCAFS LC-40 and SLC-40 sit close together in the source coordinates.
- Geography supports site comparison, but does not establish landing success.



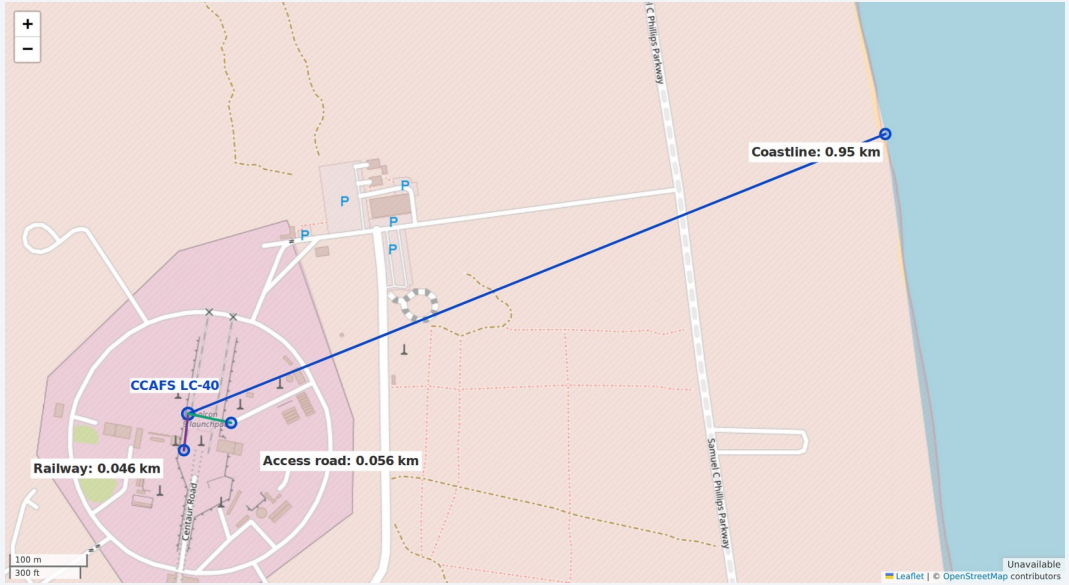
Launch Outcomes on the Map

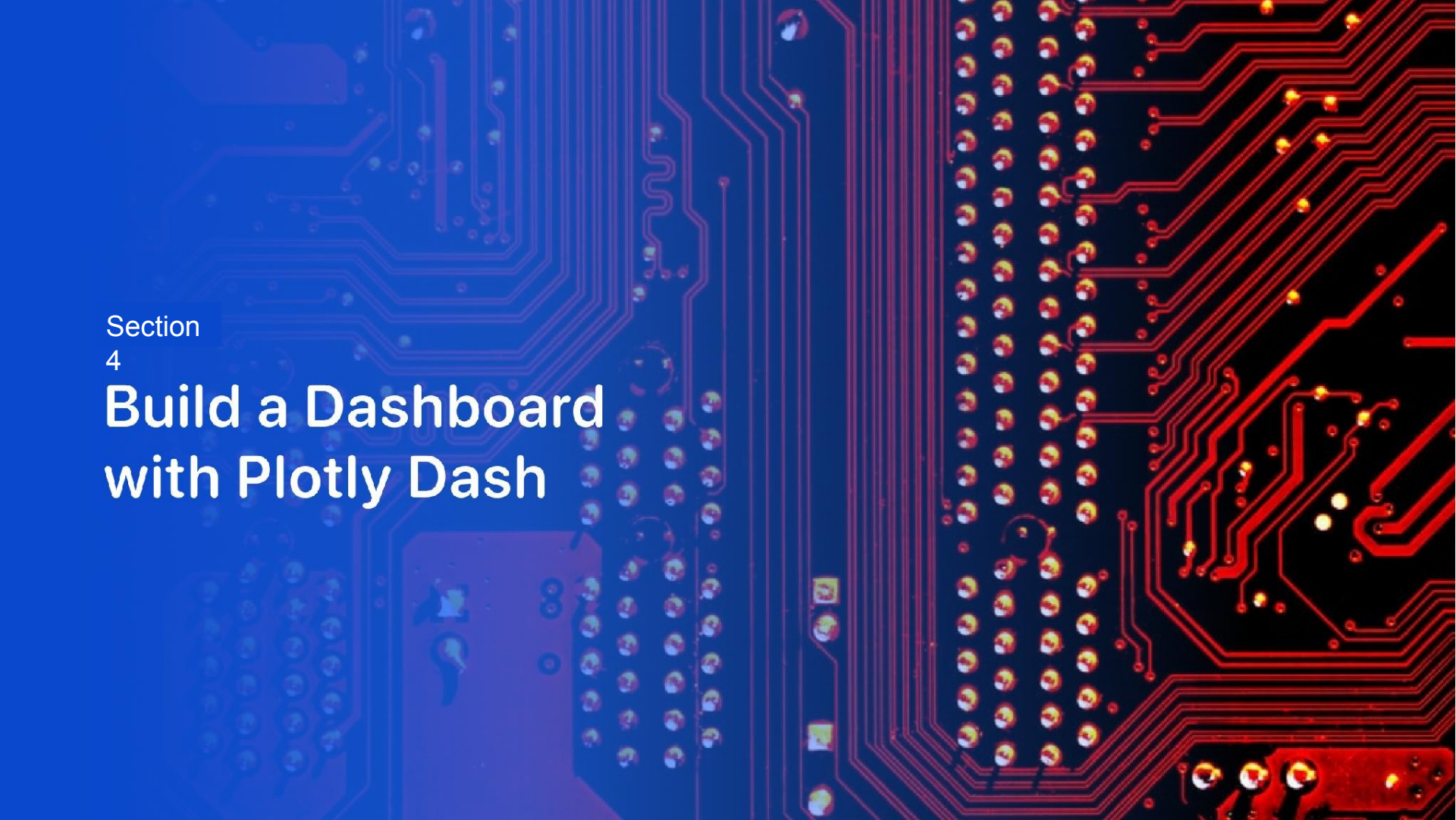
- Green points show success and red points show failure.
- Florida sites contain both outcomes.
- A small display offset exposes records that share coordinates.



CCAFS LC-40: Nearby Features

- Selected coastline point: 0.95 km.
- Selected railway node: 0.046 km.
- Selected access-road node: 0.056 km.
- Haversine distances are straight-line estimates to mapped nodes.



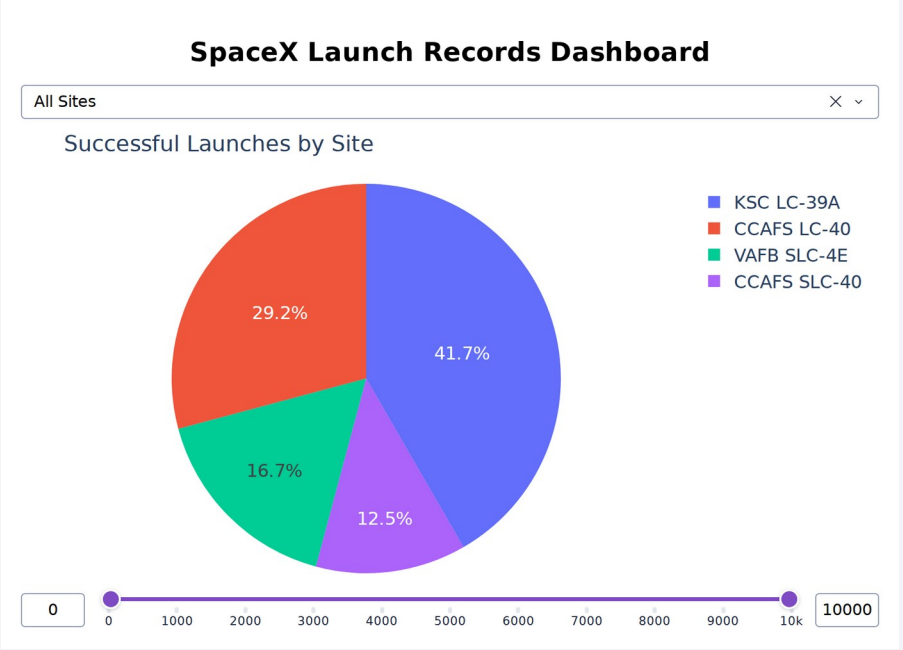


Section 4

Build a Dashboard with Plotly Dash

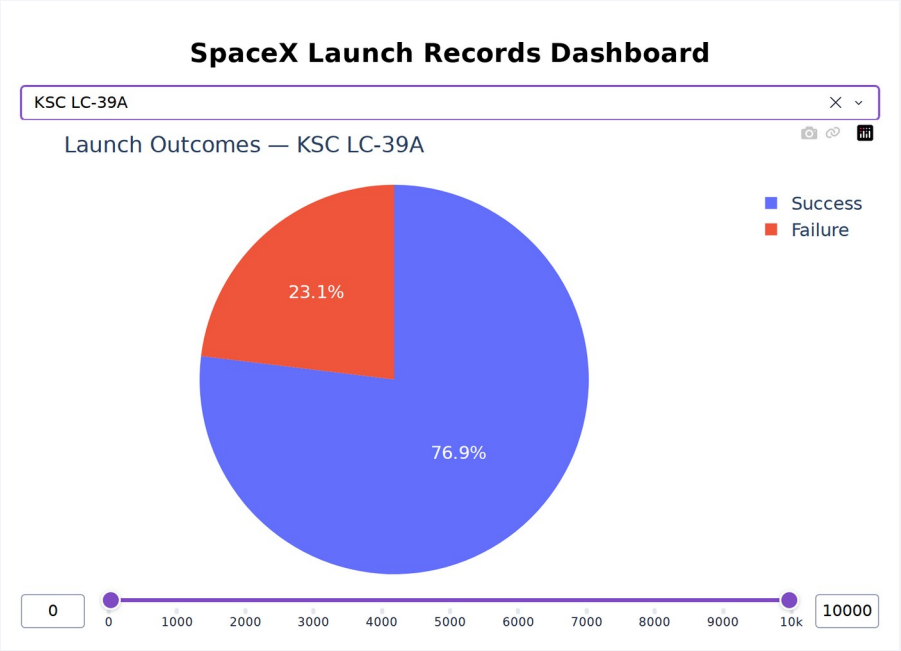
Dashboard: Success Counts by Site

- KSC LC-39A contributes 10 of the 24 successful launches.
- Its pie slice is 41.7% of all successes.
- This share compares counts, not the success rate within each site.

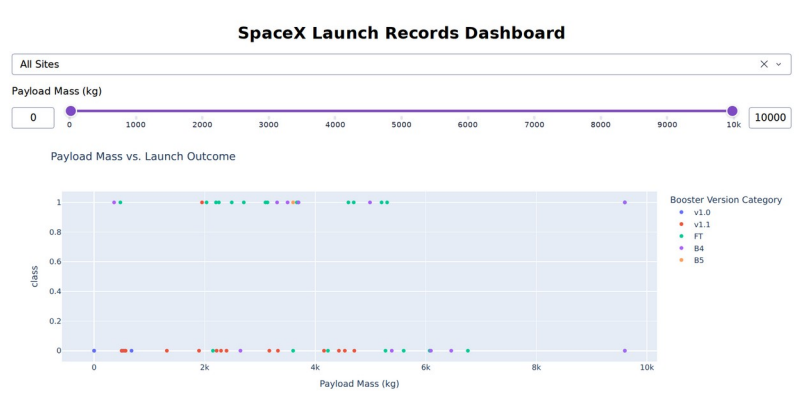


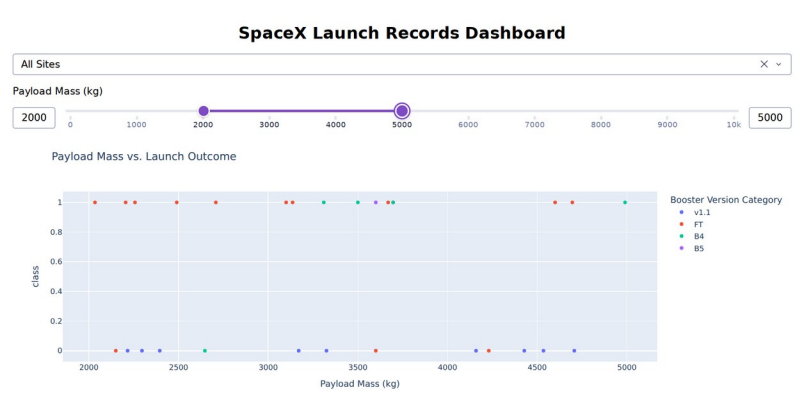
Dashboard: Highest Site Success Rate

- KSC LC-39A has 10 successes in 13 launches.
- Its 76.9% success rate is the highest among the four dashboard site labels.
- The comparison describes this 56-launch snapshot.



Dashboard: Payload Range Comparison





0-10,000 kg: 24/56 successes (42.9%).
2,000-5,000 kg: 16/29 successes (55.2%).

Success and failure overlap. The narrower range is an exploratory filter, not a causal advantage.

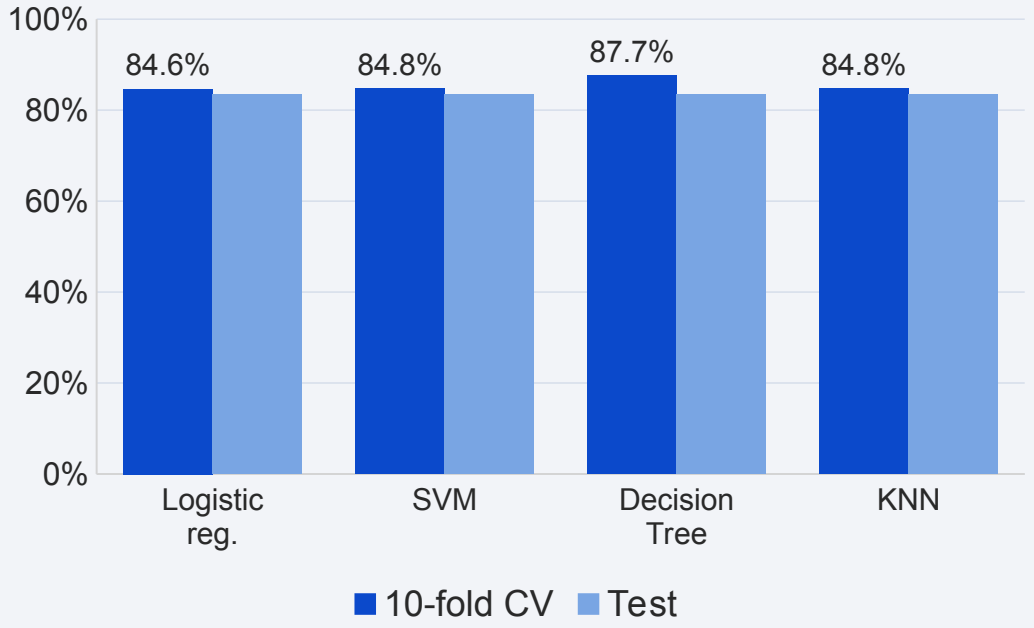
Section
5

Predictive Analysis (Classification)



Classification Accuracy

- **10-fold CV:** Logistic 84.64%, SVM 84.82%, Tree 87.68%, KNN 84.82%.
- **Test accuracy:** Each model scores 83.33%, or 15 correct predictions in 18.
- **Selection:** Decision Tree leads CV. All four share the same test predictions.



Confusion Matrix: Decision Tree

- 12 successful outcomes are correctly identified.
- 3 other outcomes are correctly rejected.
- 3 false positives predict success for an actual Class 0 outcome.
- No false negatives occur in this test sample.

Actual / predicted	Class 0	Class 1
Class 0	3	3
Class 1	0	**12**

Accuracy: 83.33%
Success precision: 80%
Success recall: 100%

Conclusions

- **Models:** Decision Tree leads CV at 87.68%. All four tie at 83.33% on 18 test launches.
- **Site insight:** KSC leads the dashboard at 76.9% (10/13). Its 41.7% share (10/24) answers a different question.
- **SQL insight:** NASA CRS payload is 45,596 kg. Including shared NASA-tagged missions gives 107,010 kg.
- **Prediction risk:** Three false positives give 80% success precision. Recall is 100% on this small test set.
- **Next step:** Fit preprocessing inside CV and test future launches chronologically before operational use.

Appendix: Data Scope and Sources

Analysis	Rows	Primary input
EDA and ML	90	dataset_part_2 / part_3
SQL	101	IBM Spacex.csv
Dashboard and map	56	IBM dash / geo CSV
Scraping output	0	Excluded from analysis

Validation priorities: fit preprocessing inside CV, test future launches, and clarify whether success means controlled landing or recoverable hardware.

https://github.com/elvenj/elvenj-ibm-data-science-professional-certificate

Thank you!



Project repository